

Part 1 - Understanding two population hypothesis testing

In statistics, we often deal with situations where we want to compare two different groups or populations. These comparisons can help us answer important questions like:

- Are students from two different schools performing differently on average in an exam?
- Do two different brands of a product have the same average quality?
- Is the recovery rate from a treatment higher in one group compared to another?

In such cases, we need to go beyond looking at a single population or sample and compare two different groups. This is where **two-population hypothesis testing** comes in handy. Instead of assuming or guessing the results, we use statistical tests to make objective decisions based on data.

At its core, **two-population hypothesis testing** allows us to determine whether there is a statistically significant difference between the means or proportions of two populations.

Let's break this down with a few simple concepts:

1. What are we comparing?

- We are comparing the **population parameters** (e.g., the means or proportions) of two groups. For example, if you want to compare the average height of men and women in a city, the **parameter** would be the average height (mean) in each group.

2. Why do we need hypothesis testing?

- Imagine you've collected data from two samples—let's say you measured the average daily coffee consumption of two departments in your office. Department A consumes 3 cups per person, and Department B consumes 3.5 cups. At first glance, this difference might seem real, but is it large enough to be statistically significant, or could it be due to random chance? Hypothesis testing helps us decide if the observed difference is due to chance or if there's a real difference between the populations.

3. Null and Alternative Hypotheses

- In any hypothesis test, we start with two competing statements:
 - The **null hypothesis (H_0)** assumes there is **no difference** between the two populations. For example, H_0 might say that the average coffee consumption is the same in both departments.
 - The **alternative hypothesis (H_1)** suggests there **is a difference**. In our example, H_1 would state that the coffee consumption differs between the two departments.

4. One-Tailed vs Two-Tailed Tests

- Sometimes, you might only be interested in whether one population is **greater than** or **less than** the other, or the difference is **greater than** or **less than** by a certain value d . In this case, you'd use a **one-tailed test**.

Eg. Test

$$H_0: \mu_1 - \mu_2 = d$$

$$H_1: \mu_1 - \mu_2 > d$$

- If you're simply checking if there's **any difference at all**, or the difference is a certain value d , without specifying the direction (higher or lower), you'd use a **two-tailed test**.

Eg. Test

$$H_0: \mu_1 - \mu_2 = d$$

$$H_1: \mu_1 - \mu_2 \neq d$$

5. Applications of Two-Population Hypothesis Testing

- Hypothesis testing for two populations is used in various fields:
 - **Business:** Comparing the average sales between two stores.
 - **Medicine:** Determining if a new drug works better than the existing one.
 - **Social Sciences:** Investigating if two demographic groups have different voting behaviors.

Part 2 – Hypothesis test on mean differences $\mu_1 - \mu_2$

Let's get started by reviewing what we've done in week 5.

Sampling distribution of sample mean:

If population follows normal distribution, sampling distribution of sample mean is _____ (**always**/sometimes/approximately) normally distributed.

If the population is non-normal (any distribution), the sampling distribution of sample mean is approximately normally distributed if the sample size is large ($n > 30$). This is called _____ (**central limit theorem**/central unlimited theorem/side limit theorem)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

This is equivalent to the standardised sample mean

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

Particularly, when the population standard deviation (variance) is unknown, the sampling distribution of the standardised sample mean follows t-distribution.

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

The Z and T statistics are called test statistics in hypothesis testing.

Expanding to two populations - Summary of your online module 9.3-9.5

Difference of two independent population mean $\mu_1 - \mu_2$			
Case	$\sigma_1^2 \sigma_2^2$ are known	$\sigma_1^2 \sigma_2^2$ are unknown and equal	$\sigma_1^2 \sigma_2^2$ are unknown and unequal
Point Estimator	$\bar{X}_1 - \bar{X}_2$		
Mean of the estimator	$\mu_1 - \mu_2$		
Standard Error of the point estimator	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$ Because two population variances are believed to be equal, we put two samples together and get a pooled estimator S_p^2 of the population variance σ^2 $S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$	$\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$ We need to find an average degree of freedom for the t distribution $df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right)^2}{\frac{1}{n_1 - 1} \left(\frac{S_1^2}{n_1} \right)^2 + \frac{1}{n_2 - 1} \left(\frac{S_2^2}{n_2} \right)^2}$
Assumptions	$n_1 \geq 30, n_2 \geq 30$ CLT; At least one $n < 30$, assume normal population		
Test Statistic	Z $= \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$	T $= \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}$	T $= \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim t_{df}$

$$n_1 - 1 + n_2 - 1$$

Example 1: Testing population mean differences assuming equal variances

Check the QBUS5001 Week 9 Lecture Data Sheet "Example 1". A study is conducted to compare the average satisfaction rates of university life between two groups of postgraduate students at Sydney University. The first group consists of students who are aware of **SUPRA** (Sydney University Postgraduate Representative Association), and the second group includes students who are not aware of SUPRA. It is hypothesised that students who are aware of SUPRA have a higher average satisfaction rate compared to those who are not aware.

Two random samples of students are taken:

- Group 1: 40 students who are aware of SUPRA
- Group 2: 35 students who are not aware of SUPRA

The satisfaction rates (on a scale from 1 to 10, where 10 represents maximum satisfaction) for both groups are recorded as follows:

- Group 1 (Aware of SUPRA):** Sample mean satisfaction = 8.32, sample standard deviation = 0.46
- Group 2 (Not aware of SUPRA):** Sample mean satisfaction = 7.46, sample standard deviation = 0.4

Assuming equal population variance, using a **5% significance level**, test if the average satisfaction rate for students who are aware of SUPRA is significantly higher than that of students who are not aware of SUPRA.

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 > 0$$

Under H_0 , as $n_1 = 40 > 30, n_2 = 35 > 30$, the test statistic is given by

$$T = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}$$

At 5% level of significance, H_0 is rejected if the observed value of the test statistic

$$t_{obs} > t_{0.05, 40+35-2}$$

Now the observed value of the test statistic is given by

$$t_{obs} = 8.5449 > 1.6660$$

The H_0 is rejected at 5% level of significance, and we have sufficient evidence that the average satisfaction score is higher in the group of students who are aware of SUPRA compared to those who are not.

Example 2: Testing population mean differences assuming unequal variances

Check the QBUS5001 Week 9 Lecture Data Sheet “Example 2”. A study is conducted to compare the average satisfaction rates of university life between two groups of postgraduate students at Sydney University. The first group consists of students who are aware of **SUPRA** (Sydney University Postgraduate Representative Association), and the second group includes students who are not aware of SUPRA. It is hypothesised that students who are aware of SUPRA have a higher average satisfaction rate compared to those who are not aware.

Two random samples of students are taken:

- Group 1: 40 students who are aware of SUPRA
- Group 2: 35 students who are not aware of SUPRA

The satisfaction rates (on a scale from 1 to 10, where 10 represents maximum satisfaction) for both groups are recorded as follows:

- **Group 1 (Aware of SUPRA):** Sample mean satisfaction = 8.32, sample standard deviation = 0.46
- **Group 2 (Not aware of SUPRA):** Sample mean satisfaction = 7.46, sample standard deviation = 0.4

Assuming unequal population variance, using a **5% significance level**, test if the average satisfaction rate for students who are aware of SUPRA is more than 0.5 points higher than that of students who are not aware of SUPRA.

$$H_0: \mu_1 - \mu_2 = 0.5$$

$$H_1: \mu_1 - \mu_2 > 0.5$$

Under H_0 , as $n_1 = 40 > 30, n_2 = 35 > 30$, the test statistic is given by

$$T = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim t_{df}$$

At 5% level of significance, H_0 is rejected if the observed value of the test statistic

$$t_{obs} > t_{0.05, 40+35-2}$$

Now the observed value of the test statistic is given by

$$t_{obs} = 3.6063 > 1.6660$$

The H_0 is rejected at 5% level of significance, and we have sufficient evidence that the average satisfaction score is more than 0.5 points higher in the group of students who are aware of SUPRA compared to those who are not.

Part 3 - Test for equal variance and normal distribution

3.1 F test for equal variance

When comparing two populations, one common assumption is that their variances (or spread of data) are equal. However, before performing certain statistical tests, like the two-sample t-test, it's important to check whether this assumption is valid. That's where the **F-test for equal variances** comes into play.

$$\mu_1 - \mu_2 = 0$$

What is the F-test?

The **F-test** is a statistical test used to determine whether two populations have equal variances. It compares the ratio of two sample variances and determines whether this ratio is significantly different from 1, which would indicate unequal variances.

$$\frac{\sigma_1^2}{\sigma_2^2} = 1$$

How does the F-test work?

The test compares the variances from two independent samples:

- **Null Hypothesis (H_0):** The variances of the two populations are equal.

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$\sigma_1^2 - \sigma_2^2 = 0$$

- **Alternative Hypothesis (H_1):** The variances of the two populations are not equal.

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

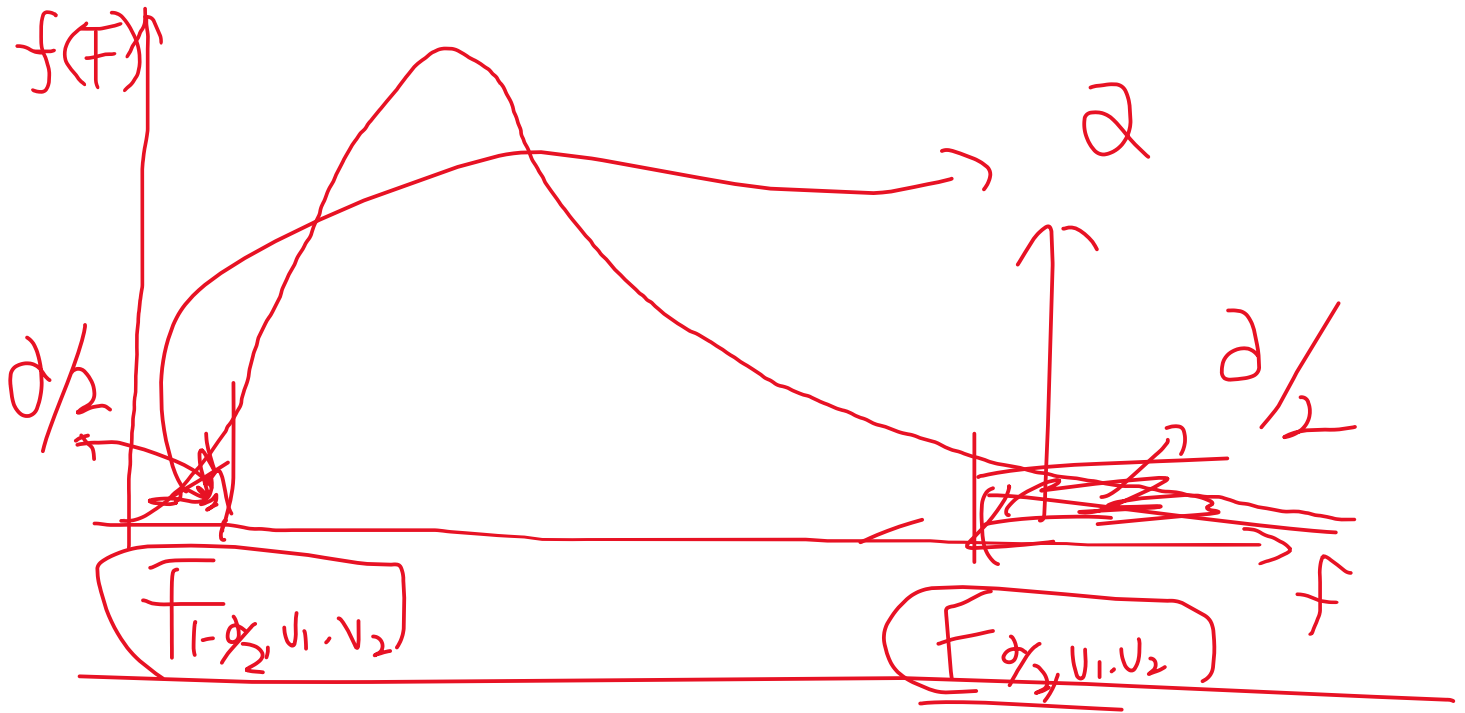
$$\frac{\sigma_1^2}{\sigma_2^2} \neq 1$$

The F-test uses the **F-ratio**:

$$F = \frac{S_1^2}{S_2^2}$$

If the calculated F-value is significantly different from 1, it indicates that the variances are not equal.

So how do we determine if the F-value is significantly different from 1?



Example 3

Check the QBUS5001 Week 9 Lecture Data Sheet "Example 3". Test if the population variances are equal between group 1 and group 2.

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

Under H_0 , the test statistic is given by

$$F = \frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1}$$

At 5% level of significance, the H_0 will be rejected if the observed F

$$F_{stat} > F_{\frac{\alpha}{2}, n_1-1, n_2-1} = F.INV$$

Or

$$F_{stat} < F_{1-\frac{\alpha}{2}, n_1-1, n_2-1}$$

Note: we usually use p-value to make the decisions instead of the rejection region

H_0 is rejected if

$$p - \text{value} < \alpha$$

Now the p-value is given by

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$$p - value = 0.4418 > 0.05$$

The H_0 is not rejected at 5% level of significance, we can conclude that the two populations have equal variances.

3.2 χ^2 test for normality

When conducting statistical analyses, one of the most common assumptions is that the data follows a **normal distribution**. Many statistical tests, like the t-test, assume normality. However, it's important to first verify whether this assumption holds for your data. One way to check this is through the **Chi-Square Test for Normality**.

What is the Chi-Square Test for Normality?

The **Chi-Square Test for Normality** is a statistical method used to determine if a set of data follows a normal distribution. It compares the observed frequency distribution of the data to the expected frequencies based on a normal distribution.

When is the Chi-Square Test for Normality Used?

This test is used in situations where you want to assess the goodness of fit between your data and the normal distribution. If the data does not fit a normal distribution, you may need to use non-parametric tests or transform the data before further analysis.

How does the Chi-Square Test for Normality Work?

The test compares the **observed** number of data points within different intervals (bins) to the **expected** number of data points, assuming the data follows a normal distribution.

Null Hypothesis (H_0): The data follows a normal distribution.

H_0 : Data follows a normal distribution.

Alternative Hypothesis (H_1): The data does not follow a normal distribution.

H_1 : Data does not follow a normal distribution.

Example 4

Please refer to the QBUS5001 Week 9 Lecture Data Sheet, "Example 4." Test whether both Group 1 and Group 2 follow a normal distribution.

Check the excel output.

Group 1 data is normal

Group 2 data is not normal

Part 4 – Test proportions for two populations

In many real-world scenarios, we often need to compare proportions between two distinct groups. This type of analysis, called **two population proportion testing**, helps us determine whether the difference between two proportions is statistically significant or just due to random variation.

What is Testing Two Population Proportions?

When we want to compare the proportion of a certain characteristic (e.g., success rates, approval rates, or defect rates) in two different populations, we use a hypothesis test for two population proportions. This type of test allows us to examine whether the observed difference between the two proportions is large enough to conclude that there is a real difference, rather than one caused by random chance.

When Do We Use This Test?

This test is used in situations like:

- **Comparing pass rates** between two different schools.
- **Comparing the proportion of voters** supporting two candidates.
- **Comparing defect rates** between two manufacturing processes.

How Does the Test Work?

The goal is to compare the proportions from two independent samples and assess whether the difference is statistically significant. We set up two competing hypotheses:

- **Null Hypothesis (H_0):** The proportions of the two populations are equal.

$$H_0: p_1 = p_2$$

- **Alternative Hypothesis (H_1):** The proportions of the two populations are not equal (two-tailed), or one proportion is larger than the other (one-tailed).

$$H_1: p_1 \neq p_2 \text{ (two-tailed) or}$$

$$H_1: p_1 > p_2 \text{ or } H_1: p_1 < p_2 \text{ (one-tailed)}$$

Two population proportions $p_1 - p_2$		
Point Estimator	$\hat{p}_1 - \hat{p}_2$	
Sampling Distribution	Normal	
mean	$p_1 - p_2$	
Standard Error	$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	
Assumptions	$n_1 p_1 \geq 5$ $n_1(1-p_1) \geq 5$; $n_2 p_2 \geq 5$ $n_2(1-p_2) \geq 5$ p_1 and p_2 are unknown, so we use $\hat{p}_1$ and $\hat{p}_2$ instead	
Test Statistic	$H_0: p_1 - p_2 = 0$	$Z = \frac{\hat{p}_1 - \hat{p}_2 - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1)$ Where $\hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$

Example 5

SUPRA (Sydney University Postgraduate Representative Association) offers free academic appeal services to assist students with their academic grievances. Some students choose to use external paid services for the same purpose. It is believed that the success rate of SUPRA's free academic appeal services is higher than that of external paid services.

To test this claim, a random sample of 150 students who used SUPRA's services is taken, and 105 of them had successful outcomes. Additionally, a random sample of 130 students who used external paid services is taken, with 78 of them having successful outcomes.

Using a 5% level of significance, test whether the success rate of SUPRA's free academic appeal services is significantly higher than that of external paid services.

$$H_0: p_1 - p_2 = 0$$

$$H_1: p_1 - p_2 > 0$$

Based on the information,

$$\hat{p}_1 = \frac{X_1}{n_1} = \frac{105}{150} = 0.7$$

$$\hat{p}_2 = \frac{X_2}{n_2} = \frac{78}{130} = 0.6$$

$$\hat{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{105 + 78}{150 + 130} = 0.6536$$

The standard error of $\hat{p}_1 - \hat{p}_2$

$$\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = 0.0570$$

Check

$$n_1 \hat{p}_1 = 105 > 5$$

$$n_1 (1 - \hat{p}_1) = 45 > 5$$

$$n_2 \hat{p}_2 = 78 > 5$$

$$n_2 (1 - \hat{p}_2) = 52 > 5$$

Under H_0 the test stat is given by

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - 0}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim N(0,1)$$

H_0 is rejected if the observed value of the test statistic

$$z_{obs} > z_{0.05} = 1.645$$

Now, the observed value of the test stat is

$$z_{obs} = \frac{\hat{p}_1 - \hat{p}_2 - 0}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.7 - 0.6}{0.0570} = 1.7538 > 1.645$$

Therefore, H_0 is rejected at 5% level of significance, and we have sufficient evidence that the success rate of SUPRA's free academic appeal service is significantly higher than that of external paid services.